Atiyah–Macdonald, Chapter 3: Rings and Modules of Fractions

Solutions to the exercises

Conventions. S is a multiplicatively closed subset of A, $f : A \to S^{-1}A$, $a \mapsto a/1$ is the canonical map, and $A_{\mathfrak{p}} = S^{-1}A$ for $S = A - \mathfrak{p}$, $A_f = S^{-1}A$ for $S = \{f^n\}_{n \geq 0}$. From the text I use: the universal property (3.1, 3.2); exactness of S^{-1} (3.3) and its consequences 3.4; $S^{-1}A \otimes_A M \cong S^{-1}M$ (3.5) and flatness of $S^{-1}A$ (3.6); $S^{-1}M \otimes_{S^{-1}A} S^{-1}N \cong S^{-1}(M \otimes N)$ (3.7); the local properties 3.8, 3.9, 3.10; the description of ideals and primes of $S^{-1}A$ (3.11: every ideal of $S^{-1}A$ is extended; $\mathfrak{a}^e = (1) \iff \mathfrak{a} \cap S \neq \emptyset$; primes of $S^{-1}A \leftrightarrow$ primes of A not meeting S), 3.12, 3.13 (primes of $A_{\mathfrak{p}} \leftrightarrow$ primes contained in $\mathfrak{p}$), 3.14 (Ann commutes with S^{-1} for finitely generated M), and 3.16. Also $S^{-1}A = 0 \iff 0 \in S$.

This chapter covers the unnumbered exercise on page 37 and the 30 end-of-chapter exercises. In Exercise 25 the formula for h as printed is not additive; I explain this there.

In-text exercise

Exercise p. 37

> **Problem.** Verify that the definitions of addition and multiplication of fractions $(a/s) + (b/t) = (at + bs)/st$, $(a/s)(b/t) = ab/st$ are independent of the choices of representatives (a, s) and (b, t), and that $S^{-1}A$ satisfies the axioms of a commutative ring with identity.

Solution. Recall $(a, s) \equiv (b, t) \iff (at - bs)u = 0$ for some $u \in S$. Suppose $a/s = a'/s'$ (witness u: $(as' - a's)u = 0$) and $b/t = b'/t'$ (witness v). Then

$$\big[(at + bs)s't' - (a't' + b's')st\big]uv = \big[(as' - a's)tt' + (bt' - b't)ss'\big]uv = 0,$$

so $(at + bs)/st = (a't' + b's')/s't'$. For products,

$$ab\,s't' - a'b'\,st = (as' - a's)\,bt' + a's\,(bt' - b't),$$

and multiplying by uv gives 0, so $ab/st = a'b'/s't'$.

Ring axioms. Addition and multiplication are commutative by the formulas. The element $0/1$ is a zero, $1/1$ an identity, and $-(a/s) = (-a)/s$. Associativity of $+$: both $(a/s + b/t) + c/u$ and $a/s + (b/t + c/u)$ equal $(atu + bsu + cst)/stu$; associativity of $\cdot$ is similar. Distributivity: $(a/s)(b/t + c/u) = a(bu + ct)/stu$ while $(a/s)(b/t) + (a/s)(c/u) = (abus + acts)/s^2tu = s\,a(bu + ct)/(s \cdot stu)$, and these agree because $sx/sy = x/y$ for $s \in S$ (take $u = 1$: $(sx\,y - x\,sy) \cdot 1 = 0$). Finally $f : A \to S^{-1}A$ is a ring homomorphism. $\square$

End-of-chapter exercises

Exercise 1

> **Problem.** Let S be a multiplicatively closed subset of a ring A, and let M be a finitely generated A-module. Prove that $S^{-1}M = 0$ if and only if there exists $s \in S$ such that $sM = 0$.

Solution. ($\Leftarrow$) $m/t = sm/st = 0$ for all $m/t \in S^{-1}M$. ($\Rightarrow$) Let $x_1, \ldots, x_n$ generate M. Since $x_i/1 = 0$ there is $s_i \in S$ with $s_i x_i = 0$. Then $s = s_1 \cdots s_n \in S$ kills every x_i, hence $sM = 0$. $\square$

Exercise 2

> **Problem.** Let $\mathfrak{a}$ be an ideal of a ring A, and let $S = 1 + \mathfrak{a}$. Show that $S^{-1}\mathfrak{a}$ is contained in the Jacobson radical of $S^{-1}A$. Use this result and Nakayama's lemma to give a proof of (2.5) which does not depend on determinants. [If $M = \mathfrak{a}M$, then $S^{-1}M = (S^{-1}\mathfrak{a})(S^{-1}M)$, hence by Nakayama we have $S^{-1}M = 0$. Now use Exercise 1.]

Solution. $S^{-1}\mathfrak{a} \subseteq \mathfrak{R}(S^{-1}A)$. Let x/s with $x \in \mathfrak{a}$, $s \in S$, and let $b/t \in S^{-1}A$ be arbitrary ($b \in A$, $t \in S$). By 1.9 it suffices to show $1 - (x/s)(b/t) = (st - xb)/st$ is a unit. Now $st \in 1 + \mathfrak{a}$ (product of elements of $1 + \mathfrak{a}$) and $xb \in \mathfrak{a}$, so $st - xb \in 1 + \mathfrak{a} = S$. Hence $(st - xb)/st$ has numerator and denominator in S, and its inverse is $st/(st - xb)$.

Proof of 2.5 (Cor.): M finitely generated, $\mathfrak{a}M = M \Rightarrow \exists x \equiv 1 \pmod{\mathfrak{a}}$ with $xM = 0$. Put $S = 1 + \mathfrak{a}$. Applying S^{-1} to $M = \mathfrak{a}M$ gives $S^{-1}M = S^{-1}(\mathfrak{a}M) = (S^{-1}\mathfrak{a})(S^{-1}M)$ (elements of $\mathfrak{a}M$ are sums $\sum a_i m_i$ and S^{-1} commutes with this). $S^{-1}M$ is a finitely generated $S^{-1}A$-module and $S^{-1}\mathfrak{a} \subseteq \mathfrak{R}(S^{-1}A)$, so Nakayama's lemma (2.6) gives $S^{-1}M = 0$. By Exercise 1 there is $x \in S = 1 + \mathfrak{a}$ with $xM = 0$. $\square$

Exercise 3

> **Problem.** Let A be a ring, let S and T be two multiplicatively closed subsets of A, and let U be the image of T in $S^{-1}A$. Show that the rings $(ST)^{-1}A$ and $U^{-1}(S^{-1}A)$ are isomorphic.

Solution. Let $g : A \to U^{-1}(S^{-1}A)$, $g(a) = (a/1)/1$. We check the three conditions of Cor. 3.2 for $ST = \{st\}$.

(i) $g(st) = ((s/1)(t/1))/1$. Here $s/1$ is a unit of $S^{-1}A$, hence of $U^{-1}(S^{-1}A)$, and $t/1 \in U$ is a unit of $U^{-1}(S^{-1}A)$. So $g(st)$ is a unit.

(ii) If $g(a) = 0$, there is $u = t/1 \in U$ with $(t/1)(a/1) = 0$ in $S^{-1}A$, i.e. $s\,ta = 0$ for some $s \in S$; and $st \in ST$.

(iii) A general element of $U^{-1}(S^{-1}A)$ is $\xi = (a/s)/(t/1)$. Then $\xi\, g(st) = \big((a/s)(st/1)\big)/(t/1) = (at/1)/(t/1) = (a/1)/1 = g(a)$ (as $(at/1) \cdot 1 - (a/1)(t/1) = 0$). Since $g(st)$ is a unit, $\xi = g(a)g(st)^{-1}$.

By Cor. 3.2 there is a unique isomorphism $(ST)^{-1}A \to U^{-1}(S^{-1}A)$ compatible with the maps from A. $\square$

Exercise 4

> **Problem.** Let $f : A \to B$ be a homomorphism of rings and let S be a multiplicatively closed subset of A. Let $T = f(S)$. Show that $S^{-1}B$ and $T^{-1}B$ are isomorphic as $S^{-1}A$-modules.

Solution. $S^{-1}B$ means the localization of B regarded as an A-module (s acts as $f(s)$). Define $\varphi : S^{-1}B \to T^{-1}B$, $b/s \mapsto b/f(s)$. For $b, b' \in B$, $s, s' \in S$:

$$b/s = b'/s' \text{ in } S^{-1}B \iff \exists t \in S :\ t(s'b - sb') = 0 \iff \exists t \in S :\ f(t)\big(f(s')b - f(s)b'\big) = 0 \iff b/f(s) = b'/f(s')$$

because every element of T has the form $f(t)$. So φ is well defined and injective; it is clearly surjective, additive, and $S^{-1}A$-linear: $(a/s_1)\cdot(b/s) = f(a)b/f(s_1 s) \mapsto f(a)b/f(s_1)f(s)$, which is the action of a/s_1 on $b/f(s)$ via the homomorphism $S^{-1}A \to T^{-1}B$ induced by f. (In fact φ is a ring isomorphism.) $\square$

Exercise 5

> **Problem.** Let A be a ring. Suppose that, for each prime ideal $\mathfrak{p}$, the local ring $A_\mathfrak{p}$ has no nilpotent element $\neq 0$. Show that A has no nilpotent element $\neq 0$. If each $A_\mathfrak{p}$ is an integral domain, is A necessarily an integral domain?

Solution. Let $x \in A$ be nilpotent. For each prime $\mathfrak{p}$, $x/1 \in A_\mathfrak{p}$ is nilpotent, hence 0, so $sx = 0$ for some $s \notin \mathfrak{p}$, i.e. $\mathrm{Ann}(x) \not\subseteq \mathfrak{p}$. Thus $\mathrm{Ann}(x)$ lies in no prime ideal, so $\mathrm{Ann}(x) = (1)$ and $x = 0$.

No, A need not be an integral domain. Let k be a field and $A = k \times k$. Its primes are $\mathfrak{p}_1 = 0 \times k$ and $\mathfrak{p}_2 = k \times 0$. For $S = A - \mathfrak{p}_1 \ni e_1 = (1,0)$ we have $e_1 e_2 = 0$ with $e_2 = (0,1)$, so $e_2/1 = 0$ in $A_{\mathfrak{p}_1}$ and $A_{\mathfrak{p}_1} \cong S^{-1}(A/(e_2)) \cong S^{-1}k = k$ (the image of S in $A/(e_2) \cong k$ consists of non-zero scalars), a field. Likewise $A_{\mathfrak{p}_2} \cong k$. So every $A_\mathfrak{p}$ is a domain, but $A = k \times k$ has zero-divisors. $\square$

Exercise 6

> **Problem.** Let A be a ring $\neq 0$ and let Σ be the set of all multiplicatively closed subsets S of A such that $0 \notin S$. Show that Σ has maximal elements, and that $S \in \Sigma$ is maximal if and only if $A - S$ is a minimal prime ideal of A.

Solution. $\Sigma \ni \{1\}$ is non-empty, and the union of a chain in Σ is multiplicatively closed and does not contain 0; by Zorn's lemma Σ has maximal elements.

First, for $S \in \Sigma$ we have $S^{-1}A \neq 0$, so it has a maximal ideal, whose contraction is a prime $\mathfrak{q}$ of A with $\mathfrak{q} \cap S = \emptyset$ (3.11(iv)).

($\Rightarrow$) Let S be maximal and $\mathfrak{q}$ a prime with $\mathfrak{q} \cap S = \emptyset$ as above. Then $A - \mathfrak{q}$ is multiplicatively closed, contains S and not 0, so $A - \mathfrak{q} \in \Sigma$ and by maximality $A - \mathfrak{q} = S$, i.e. $A - S = \mathfrak{q}$ is

prime. If $\mathfrak{q}' \subseteq \mathfrak{q}$ is a prime, then $A - \mathfrak{q}' \in \Sigma$ contains S, hence equals S, so $\mathfrak{q}' = \mathfrak{q}$. Thus $A - S$ is a minimal prime.

($\Leftarrow$) Let $\mathfrak{p}$ be a minimal prime and $S = A - \mathfrak{p}$; $S \in \Sigma$ since $0 \in \mathfrak{p}$. If $S \subseteq S' \in \Sigma$, there is a prime $\mathfrak{q}$ with $\mathfrak{q} \cap S' = \emptyset$, so $\mathfrak{q} \subseteq A - S' \subseteq A - S = \mathfrak{p}$; by minimality $\mathfrak{q} = \mathfrak{p}$, so $A - S' \supseteq \mathfrak{p} = A - S$, i.e. $S' \subseteq S$. Hence $S' = S$ and S is maximal. □

Exercise 7

Problem. A multiplicatively closed subset S of a ring A is said to be *saturated* if $xy \in S \iff x \in S$ and $y \in S$. Prove that

(i) S is saturated $\iff$ $A - S$ is a union of prime ideals.

(ii) If S is any multiplicatively closed subset of A, there is a unique smallest saturated multiplicatively closed subset $\bar{S}$ containing S, and $\bar{S}$ is the complement in A of the union of the prime ideals which do not meet S. ($\bar{S}$ is called the *saturation* of S.)

If $S = 1 + \mathfrak{a}$, where $\mathfrak{a}$ is an ideal of A, find $\bar{S}$.

Solution. (i) ($\Leftarrow$) If $A - S = \bigcup \mathfrak{p}_i$, then $S = \bigcap (A - \mathfrak{p}_i)$ and each $A - \mathfrak{p}_i$ is multiplicatively closed and saturated ($xy \notin \mathfrak{p}_i \iff x, y \notin \mathfrak{p}_i$, as $\mathfrak{p}_i$ is prime). An intersection of saturated multiplicatively closed sets is such a set.

($\Rightarrow$) Let S be saturated and $x \notin S$. Then $(x) \cap S = \emptyset$: if $ax \in S$ then $x \in S$. By 3.11(ii) $S^{-1}(x) \neq (1)$, so it lies in a maximal ideal of $S^{-1}A$ whose contraction $\mathfrak{p}$ is a prime with $x \in \mathfrak{p}$, $\mathfrak{p} \cap S = \emptyset$ (3.11(iv)). Hence $A - S \subseteq \bigcup \{\mathfrak{p} \text{ prime} : \mathfrak{p} \cap S = \emptyset\} \subseteq A - S$, so $A - S$ is a union of primes.

(ii) Put $\bar{S} = A - \bigcup \{\mathfrak{p} \text{ prime} : \mathfrak{p} \cap S = \emptyset\}$. By (i) it is saturated; clearly $S \subseteq \bar{S}$. If $T \supseteq S$ is saturated, then $A - T = \bigcup \mathfrak{q}$ with $\mathfrak{q} \cap T = \emptyset$ (by the proof of (i)), so $\mathfrak{q} \cap S = \emptyset$ and $\mathfrak{q} \subseteq A - \bar{S}$. Hence $A - T \subseteq A - \bar{S}$, i.e. $\bar{S} \subseteq T$. So $\bar{S}$ is the unique smallest saturated set containing S.

The case $S = 1 + \mathfrak{a}$: $\bar{S} = \{x \in A : \ x + \mathfrak{a}$ is a unit of $A/\mathfrak{a}\}$. Indeed, let S' be this set; it is saturated (a product is a unit mod $\mathfrak{a}$ iff each factor is) and contains $1 + \mathfrak{a}$, so $\bar{S} \subseteq S'$. Conversely if $x \in S'$ with $xy \in 1 + \mathfrak{a}$ and $x \in \mathfrak{p}$ for a prime $\mathfrak{p}$ disjoint from $1 + \mathfrak{a}$, then $xy \in \mathfrak{p} \cap (1 + \mathfrak{a})$, a contradiction; so $x \in \bar{S}$. (For $A = \mathbf{Z}$, $\mathfrak{a} = (6)$: $\bar{S} = \{x : \gcd(x, 6) = 1\}$.) □

Exercise 8

Problem. Let S, T be multiplicatively closed subsets of A, such that $S \subseteq T$. Let $\phi : S^{-1}A \to T^{-1}A$ be the homomorphism which maps each $a/s \in S^{-1}A$ to a/s considered as an element of $T^{-1}A$. Show that the following statements are equivalent:

(i) ϕ is bijective.

(ii) For each $t \in T$, $t/1$ is a unit in $S^{-1}A$.

(iii) For each $t \in T$ there exists $x \in A$ such that $xt \in S$.

(iv) T is contained in the saturation of S (Exercise 7).

(v) Every prime ideal which meets T also meets S.

Solution. (i)⇒(ii): $t/1$ is a unit of $T^{-1}A$ and $\phi(t/1) = t/1$; ϕ^{-1} is a ring homomorphism, so $t/1$ is a unit of $S^{-1}A$.

(ii)⇒(i): By 3.1 applied to $A \to S^{-1}A$ (which sends T to units) there is a homomorphism $\psi : T^{-1}A \to S^{-1}A$, $\psi(a/t) = (a/1)(t/1)^{-1}$. Then $\psi\phi(a/s) = (a/1)(s/1)^{-1} = a/s$ and $\phi\psi(a/t) = a/t$ in $T^{-1}A$. So ϕ is bijective.

(ii)⇔(iii): $t/1$ is a unit in $S^{-1}A$ iff $(t/1)(x/s) = 1$ for some $x \in A$, $s \in S$, iff $u(xt - s) = 0$ for some $u \in S$. Then $(ux)t = us \in S$. Conversely if $x't = s \in S$ then $(t/1)(x'/s) = 1$.

(iii)⇒(iv): if $xt \in S \subseteq \bar{S}$ and $\bar{S}$ is saturated, then $t \in \bar{S}$.

(iv)⇒(iii): Let $t \in T \subseteq \bar{S}$. If $(t) \cap S = \emptyset$, then as in Exercise 7 there is a prime $\mathfrak{p} \ni t$ with $\mathfrak{p} \cap S = \emptyset$, so $t \notin \bar{S}$, a contradiction. So $xt \in S$ for some x.

(iv)⇒(v): if a prime $\mathfrak{p}$ contains $t \in T \subseteq \bar{S}$, then by definition of $\bar{S}$, $\mathfrak{p}$ meets S.

(v)⇒(iv): if $t \in T$ were not in $\bar{S}$, then $t \in \mathfrak{p}$ for a prime $\mathfrak{p}$ disjoint from S; $\mathfrak{p}$ meets T, contradicting (v). □

Exercise 9

Problem. The set S_0 of all non-zero-divisors in A is a saturated multiplicatively closed subset of A. Hence the set D of zero-divisors in A is a union of prime ideals (see Chapter 1, Exercise 14). Show that every minimal prime ideal of A is contained in D. [Use Exercise 6.]

The ring $S_0^{-1}A$ is called the *total ring of fractions* of A. Prove that

(i) S_0 is the largest multiplicatively closed subset of A for which the homomorphism $A \to S^{-1}A$ is injective.

(ii) Every element in $S_0^{-1}A$ is either a zero-divisor or a unit.

(iii) Every ring in which every non-unit is a zero-divisor is equal to its total ring of fractions (i.e., $A \to S_0^{-1}A$ is bijective).

Solution. *Minimal primes lie in D.* Let $\mathfrak{p}$ be a minimal prime and $x \in \mathfrak{p}$. By Exercise 6, $A - \mathfrak{p}$ is a maximal element of Σ; equivalently $A_\mathfrak{p}$ has exactly one prime ideal (3.13), so $\mathfrak{p}A_\mathfrak{p}$ is the nilradical of $A_\mathfrak{p}$ and $x/1$ is nilpotent: $sx^n = 0$ for some $s \notin \mathfrak{p}$ and $n \geq 1$ (note $s \neq 0$ as $0 \in \mathfrak{p}$). Take n minimal with $sx^n = 0$; then $y = sx^{n-1} \neq 0$ ($n = 1$: $y = s \neq 0$) and $xy = 0$. So $x \in D$.

(i) $f : A \to S^{-1}A$ is injective iff ($sa = 0$ with $s \in S \Rightarrow a = 0$), i.e. iff no $s \in S$ is a zero-divisor, i.e. iff $S \subseteq S_0$.

(ii) Let $a/s \in S_0^{-1}A$. If $a \in S_0$, a/s is a unit with inverse s/a. If $a \notin S_0$, then a is a zero-divisor: $ab = 0$ with $b \neq 0$. By (i) $b/1 \neq 0$, and $(a/s)(b/1) = 0$, so a/s is a zero-divisor.

(iii) If every non-unit is a zero-divisor then $S_0 \subseteq A^*$ (units). Then $a/s = (as^{-1})/1$, so f is surjective; it is injective by (i). □

Exercise 10

Problem. Let A be a ring.

(i) If A is absolutely flat (Chapter 2, Exercise 27) and S is any multiplicatively closed subset of A, then $S^{-1}A$ is absolutely flat.

(ii) A is absolutely flat $\iff$ $A_{\mathfrak{m}}$ is a field for each maximal ideal $\mathfrak{m}$.

Solution. (i) By Ch. 2, Ex. 27, A is absolutely flat iff $x = ax^2$ is solvable for each x. Given $\xi = x/s \in S^{-1}A$ pick a with $x = ax^2$. Then $\xi^2 \cdot (as/1) = ax^2s/s^2 = x/s = \xi$, so $\xi \in (\xi^2)$, i.e. $(\xi) = (\xi^2)$. By Ch. 2, Ex. 27 again, $S^{-1}A$ is absolutely flat.

(ii) ($\Rightarrow$) $A_{\mathfrak{m}}$ is absolutely flat by (i) and local, hence a field by Ch. 2, Ex. 28. ($\Leftarrow$) Let $x \in A$ and $M = (x)/(x^2)$. For each maximal $\mathfrak{m}$, $M_{\mathfrak{m}} = (x/1)A_{\mathfrak{m}}/(x^2/1)A_{\mathfrak{m}}$ (3.3, 3.4) and in the field $A_{\mathfrak{m}}$ the ideals $(x/1)$ and $(x/1)^2 = (x^2/1)$ coincide (both are 0 or the whole field). So $M_{\mathfrak{m}} = 0$ for all $\mathfrak{m}$, hence $M = 0$ by 3.8, i.e. $(x) = (x^2)$. By Ch. 2, Ex. 27, A is absolutely flat. □

Exercise 11

Problem. Let A be a ring. Prove that the following are equivalent:

(i) $A/\mathfrak{N}$ is absolutely flat ($\mathfrak{N}$ being the nilradical of A).

(ii) Every prime ideal of A is maximal.

(iii) $\operatorname{Spec}(A)$ is a T_1-space (i.e., every subset consisting of a single point is closed).

(iv) $\operatorname{Spec}(A)$ is Hausdorff.

If these conditions are satisfied, show that $\operatorname{Spec}(A)$ is compact and totally disconnected (i.e. the only connected subsets of $\operatorname{Spec}(A)$ are those consisting of a single point).

Solution. Put $\bar{A} = A/\mathfrak{N}$. Primes of A correspond to primes of $\bar{A}$ (all primes contain $\mathfrak{N}$), maximal to maximal, and $\operatorname{Spec} A \cong \operatorname{Spec} \bar{A}$ (Ch. 1, Ex. 21(iv)).

(ii)$\Leftrightarrow$(iii): by Ch. 1, Ex. 18(i), $\{x\}$ is closed iff $\mathfrak{p}_x$ is maximal.

(i)$\Rightarrow$(ii): Let $B = \bar{A}$ be absolutely flat, $\mathfrak{q}$ a prime of B, $\mathfrak{m} \supseteq \mathfrak{q}$ a maximal ideal. By Exercise 10(ii), $B_{\mathfrak{m}}$ is a field, so it has exactly one prime ideal; by 3.13 the primes of B contained in $\mathfrak{m}$ correspond to the primes of $B_{\mathfrak{m}}$, so $\mathfrak{m}$ is the only prime contained in $\mathfrak{m}$. Hence $\mathfrak{q} = \mathfrak{m}$.

(ii)$\Rightarrow$(i): Let all primes of $\bar{A}$ be maximal. For a maximal $\mathfrak{m}$ of $\bar{A}$, the local ring $\bar{A}_{\mathfrak{m}}$ has a unique prime (3.13, since every prime inside $\mathfrak{m}$ is $\mathfrak{m}$), so $\bar{A}_{\mathfrak{m}}/\mathfrak{N}(\bar{A}_{\mathfrak{m}})$ is a field (Ch. 1, Ex. 10). By 3.12, $\mathfrak{N}(\bar{A}_{\mathfrak{m}}) = (\mathfrak{N}\bar{A})_{\mathfrak{m}} = 0$. So $\bar{A}_{\mathfrak{m}}$ is a field for all $\mathfrak{m}$ and $\bar{A}$ is absolutely flat by Exercise 10(ii).

(iv)$\Rightarrow$(iii): Hausdorff implies T_1.

(i)–(iii)$\Rightarrow$(iv): Replace A by the absolutely flat $\bar{A}$ (same spectrum). Let $\mathfrak{p} \neq \mathfrak{q}$ be primes; both are maximal, so there is $x \in \mathfrak{p} - \mathfrak{q}$. By Ch. 2, Ex. 27, $(x) = (e)$ with e idempotent. Then $e \in \mathfrak{p}$ and $e \notin \mathfrak{q}$ (else $x \in (e) \subseteq \mathfrak{q}$). From $e(1-e) = 0$ we get $1 - e \in \mathfrak{q}$ and $1 - e \notin \mathfrak{p}$. Thus $\mathfrak{q} \in X_e$, $\mathfrak{p} \in X_{1-e}$ and $X_e \cap X_{1-e} = X_{e(1-e)} = X_0 = \emptyset$ (Ch. 1, Ex. 17). So $\operatorname{Spec} A$ is Hausdorff.

Compact and totally disconnected. $\operatorname{Spec} A$ is quasi-compact (Ch. 1, Ex. 17(v)) and Hausdorff, hence compact. For $\mathfrak{p} \neq \mathfrak{q}$ and e as above, $X = X_e \sqcup X_{1-e}$: any prime contains exactly one of $e, 1-e$ (their product is 0, their sum 1). So X_e, X_{1-e} are disjoint open sets covering X, with $\mathfrak{q} \in X_e$, $\mathfrak{p} \in X_{1-e}$. A connected subset cannot contain both $\mathfrak{p}$ and $\mathfrak{q}$, so connected subsets are single points. □

Exercise 12

Problem. Let A be an integral domain and M an A-module. An element $x \in M$ is a *torsion element* of M if $\mathrm{Ann}(x) \neq 0$, that is if x is killed by some non-zero element of A. Show that the torsion elements of M form a submodule of M. This submodule is called the *torsion submodule* of M and is denoted by $T(M)$. If $T(M) = 0$, the module M is said to be torsion-free. Show that

(i) If M is any A-module, then $M/T(M)$ is torsion-free.

(ii) If $f : M \to N$ is a module homomorphism, then $f(T(M)) \subseteq T(N)$.

(iii) If $0 \to M' \to M \to M''$ is an exact sequence, then the sequence $0 \to T(M') \to T(M) \to T(M'')$ is exact.

(iv) If M is any A-module, then $T(M)$ is the kernel of the mapping $x \mapsto 1 \otimes x$ of M into $K \otimes_A M$, where K is the field of fractions of A.

[For (iv), show that K may be regarded as the direct limit of its submodules $A\xi$ ($\xi \in K$); using Chapter 1, Exercise 15 and Exercise 20, show that if $1 \otimes x = 0$ in $K \otimes M$ then $1 \otimes x = 0$ in $A\xi \otimes M$ for some $\xi \neq 0$. Deduce that $\xi^{-1}x = 0$.]

Solution. *$T(M)$ is a submodule.* $0 \in T(M)$ since $\mathrm{Ann}(0) = A \neq 0$. If $ax = 0$, $by = 0$ with $a, b \neq 0$, then $ab \neq 0$ (A a domain) and $ab(x + y) = 0$; and $a(cx) = 0$ for $c \in A$.

(i) Suppose $a\bar{x} = 0$ in $M/T(M)$ with $a \neq 0$. Then $ax \in T(M)$, so $bax = 0$ for some $b \neq 0$; $ba \neq 0$, so $x \in T(M)$, $\bar{x} = 0$.

(ii) If $ax = 0$ with $a \neq 0$ then $af(x) = 0$.

(iii) Let $u : M' \to M$, $v : M \to M''$ be the maps; by (ii) they restrict to $T(\cdot)$. The restriction of u is injective. $vu = 0$ gives $v(u(T(M'))) = 0$. If $x \in T(M)$ with $v(x) = 0$, then $x = u(x')$ for some $x' \in M'$; if $ax = 0$ ($a \neq 0$) then $u(ax') = 0$ so $ax' = 0$ (u injective), i.e. $x' \in T(M')$.

(iv) Let $S = A - \{0\}$. Then $K = S^{-1}A$ and $K \otimes_A M \cong S^{-1}M$ by 3.5, with $1 \otimes x \mapsto x/1$. Now $x/1 = 0$ in $S^{-1}M$ iff $sx = 0$ for some $s \in S$ iff $x \in T(M)$. □

Exercise 13

Problem. Let S be a multiplicatively closed subset of an integral domain A. In the notation of Exercise 12, show that $T(S^{-1}M) = S^{-1}(TM)$. Deduce that the following are equivalent:

(i) M is torsion-free.

(ii) $M_{\mathfrak{p}}$ is torsion-free for all prime ideals $\mathfrak{p}$.

(iii) $M_{\mathfrak{m}}$ is torsion-free for all maximal ideals $\mathfrak{m}$.

Solution. Assume $0 \notin S$ (otherwise both sides are 0); then $S^{-1}A \subseteq K$ is a domain. By 3.3, $S^{-1}(TM) \subseteq S^{-1}M$.

($\supseteq$) If $m \in T(M)$, $am = 0$ with $a \neq 0$, then $(a/1)(m/s) = 0$ and $a/1 \neq 0$, so $m/s \in T(S^{-1}M)$.

($\subseteq$) If m/s is torsion, $(a/t)(m/s) = 0$ with $a/t \neq 0$ (so $a \neq 0$). Then $uam = 0$ for some $u \in S$ and $ua \neq 0$, so $m \in T(M)$ and $m/s \in S^{-1}T(M)$.

(i)⇒(ii): $T(M_\mathfrak{p}) = (TM)_\mathfrak{p} = 0$. (ii)⇒(iii): trivial. (iii)⇒(i): $(TM)_\mathfrak{m} = T(M_\mathfrak{m}) = 0$ for all $\mathfrak{m}$, so $TM = 0$ by 3.8. □

Exercise 14

Problem. Let M be an A-module and $\mathfrak{a}$ an ideal of A. Suppose that $M_\mathfrak{m} = 0$ for all maximal ideals $\mathfrak{m} \supseteq \mathfrak{a}$. Prove that $M = \mathfrak{a}M$. [Pass to the $A/\mathfrak{a}$-module $M/\mathfrak{a}M$ and use (3.8).]

Solution. Let $N = M/\mathfrak{a}M$, so $\mathfrak{a}N = 0$. For a maximal ideal $\mathfrak{m} \supseteq \mathfrak{a}$: $N_\mathfrak{m}$ is a quotient of $M_\mathfrak{m} = 0$ (3.3), so $N_\mathfrak{m} = 0$. For a maximal $\mathfrak{m} \not\supseteq \mathfrak{a}$: pick $s \in \mathfrak{a} - \mathfrak{m}$; then $sN = 0$, so $n/t = sn/st = 0$ for all $n/t \in N_\mathfrak{m}$, i.e. $N_\mathfrak{m} = 0$. Thus $N_\mathfrak{m} = 0$ for every maximal $\mathfrak{m}$ of A, hence $N = 0$ by 3.8, i.e. $M = \mathfrak{a}M$. □

Exercise 15

Problem. Let A be a ring, and let F be the A-module A^n. Show that every set of n generators of F is a basis of F. [Let $x_1, \ldots, x_n$ be a set of generators and $e_1, \ldots, e_n$ the canonical basis of F. Define $\phi : F \to F$ by $\phi(e_i) = x_i$. Then ϕ is surjective and we have to prove that it is an isomorphism. By (3.9) we may assume that A is a local ring. Let N be the kernel of ϕ and let $k = A/\mathfrak{m}$ be the residue field of A. Since F is a flat A-module, the exact sequence $0 \to N \to F \xrightarrow{\phi} F \to 0$ gives an exact sequence $0 \longrightarrow k \otimes N \longrightarrow k \otimes F \xrightarrow{1\otimes\phi} k \otimes F \longrightarrow 0$. Now $k \otimes F = k^n$ is an n-dimensional vector space over k; $1 \otimes \phi$ is surjective, hence bijective, hence $k \otimes N = 0$. Also N is finitely generated, by Chapter 2, Exercise 12, hence $N = 0$ by Nakayama's lemma. Hence ϕ is an isomorphism.] Deduce that every set of generators of F has at least n elements.

Solution. Let $x_1, \ldots, x_n$ generate $F = A^n$ and $\phi : F \to F$, $\phi(e_i) = x_i$; ϕ is surjective. The x_i form a basis iff ϕ is an isomorphism, i.e. injective. By 3.9, ϕ is injective iff $\phi_\mathfrak{m} : F_\mathfrak{m} \to F_\mathfrak{m}$ is injective for all maximal $\mathfrak{m}$; $\phi_\mathfrak{m}$ is again surjective and $F_\mathfrak{m} \cong A_\mathfrak{m}^n$, so we may assume A is local with maximal ideal $\mathfrak{m}$ and residue field k.

Let $N = \operatorname{Ker}\phi$. Since F is free, ϕ splits: there is $\sigma : F \to F$ with $\phi\sigma = 1$, and $F = N \oplus \sigma(F)$. Tensoring with k: $k \otimes F \cong (k \otimes N) \oplus (k \otimes \sigma F)$, and $1 \otimes \phi : k^n \to k^n$ is surjective, hence bijective (equal finite dimension), with kernel $k \otimes N$ (the splitting makes the sequence $0 \to k \otimes N \to k \otimes F \to k \otimes F \to 0$ exact). So $k \otimes N = N/\mathfrak{m}N = 0$. N is finitely generated (Ch. 2, Ex. 12), so $N = 0$ by Nakayama. Hence ϕ is an isomorphism.

Consequence. Let $A \neq 0$ and suppose $y_1, \ldots, y_r$ generate F with $r < n$. Then $y_1, \ldots, y_r, 0, \ldots, 0$ (n elements) generate F, so form a basis; but a basis cannot contain 0 (since $1 \cdot 0 = 0$ is a non-trivial relation). Contradiction; so $r \geq n$. □

Exercise 16

Problem. Let B be a flat A-algebra. Then the following conditions are equivalent:

(i) $\mathfrak{a}^{ec} = \mathfrak{a}$ for all ideals $\mathfrak{a}$ of A.

(ii) $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective.

(iii) For every maximal ideal $\mathfrak{m}$ of A we have $\mathfrak{m}^e \neq (1)$.

(iv) If M is any non-zero A-module, then $M_B \neq 0$.

(v) For every A-module M, the mapping $x \mapsto 1 \otimes x$ of M into M_B is injective.

(B is said to be *faithfully flat* over A.) [For (i)⇒(ii), use (3.16). (ii)⇒(iii) is clear. (iii)⇒(iv): Let x be a non-zero element of M and let $M' = Ax$. Since B is flat over A it is enough to show that $M'_B \neq 0$. We have $M' \cong A/\mathfrak{a}$ for some ideal $\mathfrak{a} \neq (1)$, hence $M'_B \cong B/\mathfrak{a}^e$. Now $\mathfrak{a} \subseteq \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$, hence $\mathfrak{a}^e \subseteq \mathfrak{m}^e \neq (1)$. Hence $M'_B \neq 0$. (iv)⇒(v): Let M' be the kernel of $M \to M_B$. Since B is flat over A, the sequence $0 \to M'_B \to M_B \to (M_B)_B$ is exact. But (Chapter 2, Exercise 13, with $N = M_B$) the mapping $M_B \to (M_B)_B$ is injective, hence $M'_B = 0$ and therefore $M' = 0$. (v)⇒(i): Take $M = A/\mathfrak{a}$.]

Solution. Let $f : A \to B$ be the structure map.

(i)⇒(ii). If $\mathfrak{p}$ is a prime of A, then $\mathfrak{p}^{ec} = \mathfrak{p}$, so by 3.16 $\mathfrak{p}$ is the contraction of a prime of B.

(ii)⇒(iii). If $\mathfrak{m} = \mathfrak{q}^c$ for a prime $\mathfrak{q}$ of B, then $\mathfrak{m}^e = (\mathfrak{q}^c)^e \subseteq \mathfrak{q} \neq (1)$.

(iii)⇒(iv). Let $0 \neq x \in M$, $M' = Ax \cong A/\mathfrak{a}$ with $\mathfrak{a} = \mathrm{Ann}(x) \neq (1)$. As B is flat, $M'_B \to M_B$ is injective, so it suffices to show $M'_B \neq 0$. By Ch. 2, Ex. 2, $M'_B \cong B \otimes A/\mathfrak{a} \cong B/\mathfrak{a}^e$. Choose a maximal $\mathfrak{m} \supseteq \mathfrak{a}$; then $\mathfrak{a}^e \subseteq \mathfrak{m}^e \neq (1)$, so $B/\mathfrak{a}^e \neq 0$.

(iv)⇒(v). Let $M' = \mathrm{Ker}(M \to M_B)$, $x \mapsto 1 \otimes x$. Since B is flat, $0 \to M'_B \to M_B \to (M_B)_B$ is exact, where $M_B \to (M_B)_B$ is $b \otimes x \mapsto b \otimes (1 \otimes x)$, i.e. $b \otimes 1 \otimes x$. This map equals $\tau \circ g$, where $g : N \to N_B$ is the injection $y \mapsto 1 \otimes y$ of Ch. 2, Ex. 13 (with $N = M_B$), and τ is the isomorphism of $B \otimes_A B \otimes_A M$ exchanging the first two factors. So it is injective, hence $M'_B = 0$, hence $M' = 0$ by (iv).

(v)⇒(i). Take $M = A/\mathfrak{a}$. The map $A/\mathfrak{a} \to B \otimes A/\mathfrak{a} = B/\mathfrak{a}^e$, $\bar{x} \mapsto \overline{f(x)}$, has kernel $\mathfrak{a}^{ec}/\mathfrak{a}$. It is injective, so $\mathfrak{a}^{ec} = \mathfrak{a}$. □

Exercise 17

Problem. Let $A \xrightarrow{f} B \xrightarrow{g} C$ be ring homomorphisms. If $g \circ f$ is flat and g is faithfully flat, then f is flat.

Solution. Let $u : M' \to M$ be an injective homomorphism of A-modules; we must show $1 \otimes u : N' \to N$ is injective, where $N' = B \otimes_A M'$, $N = B \otimes_A M$ (this map is B-linear). Consider

$$N' \longrightarrow N \longrightarrow C \otimes_B N, \qquad N' \longrightarrow C \otimes_B N' \longrightarrow C \otimes_B N,$$

which agree (both send $y \mapsto 1 \otimes (1 \otimes u)(y)$). The map $N' \to C \otimes_B N'$ is injective because g is faithfully flat (Ex. 16(v)). By 2.15 and 2.14, $C \otimes_B N' = C \otimes_B (B \otimes_A M') \cong C \otimes_A M'$ (with A acting through gf) naturally, and similarly for M; so the map $C \otimes_B N' \to C \otimes_B N$ is identified with $1 \otimes u : C \otimes_A M' \to C \otimes_A M$, injective because gf is flat. Thus the composite $N' \to C \otimes_B N$ is injective, which forces $N' \to N$ to be injective. Hence B is a flat A-algebra, i.e. f is flat. □

Exercise 18

Problem. Let $f : A \to B$ be a flat homomorphism of rings, let $\mathfrak{q}$ be a prime ideal of B and let $\mathfrak{p} = \mathfrak{q}^c$. Then $f^* : \mathrm{Spec}(B_\mathfrak{q}) \to \mathrm{Spec}(A_\mathfrak{p})$ is surjective. [For $B_\mathfrak{p}$ is flat over $A_\mathfrak{p}$ by (3.10), and $B_\mathfrak{q}$ is a local ring of $B_\mathfrak{p}$, hence is flat over $B_\mathfrak{p}$. Hence $B_\mathfrak{q}$ is flat over $A_\mathfrak{p}$ and satisfies condition (3) of Exercise 16.]

Solution. Let $S = A - \mathfrak{p}$. Since $\mathfrak{q}^c = \mathfrak{p}$, $f(S) \cap \mathfrak{q} = \emptyset$, so $B_\mathfrak{q}$ is a localization of $B_\mathfrak{p} = S^{-1}B$ (at the prime $\mathfrak{q}B_\mathfrak{p}$) and $A_\mathfrak{p} \to B_\mathfrak{q}$ is defined.

$B_\mathfrak{p} \cong A_\mathfrak{p} \otimes_A B$ is a flat $A_\mathfrak{p}$-algebra (3.10 applied to the flat A-module B; or Prop. 2.20, base change of a flat module). $B_\mathfrak{q}$ is a flat $B_\mathfrak{p}$-module (3.6, a localization), so by Ch. 2, Ex. 8(ii), $B_\mathfrak{q}$ is a flat $A_\mathfrak{p}$-algebra. The maximal ideal of $A_\mathfrak{p}$ is $\mathfrak{p}A_\mathfrak{p}$, and its extension to $B_\mathfrak{q}$ is contained in $\mathfrak{q}B_\mathfrak{q} \neq (1)$ since $f(\mathfrak{p}) \subseteq \mathfrak{q}$. So condition (iii) of Ex. 16 holds for the flat $A_\mathfrak{p}$-algebra $B_\mathfrak{q}$, hence (Ex. 16) $\mathrm{Spec}(B_\mathfrak{q}) \to \mathrm{Spec}(A_\mathfrak{p})$ is surjective. □

Exercise 19

Problem. Let A be a ring, M an A-module. The *support* of M is defined to be the set $\mathrm{Supp}(M)$ of prime ideals $\mathfrak{p}$ of A such that $M_\mathfrak{p} \neq 0$. Prove the following results:

(i) $M \neq 0 \iff \mathrm{Supp}(M) \neq \emptyset$.

(ii) $V(\mathfrak{a}) = \mathrm{Supp}(A/\mathfrak{a})$.

(iii) If $0 \to M' \to M \to M'' \to 0$ is an exact sequence, then $\mathrm{Supp}(M) = \mathrm{Supp}(M') \cup \mathrm{Supp}(M'')$.

(iv) If $M = \sum M_i$, then $\mathrm{Supp}(M) = \bigcup \mathrm{Supp}(M_i)$.

(v) If M is finitely generated, then $\mathrm{Supp}(M) = V(\mathrm{Ann}(M))$ (and is therefore a closed subset of $\mathrm{Spec}(A)$).

(vi) If M, N are finitely generated, then $\mathrm{Supp}(M \otimes_A N) = \mathrm{Supp}(M) \cap \mathrm{Supp}(N)$. [Use Chapter 2, Exercise 3.]

(vii) If M is finitely generated and $\mathfrak{a}$ is an ideal of A, then $\mathrm{Supp}(M/\mathfrak{a}M) = V(\mathfrak{a} + \mathrm{Ann}(M))$.

(viii) If $f : A \to B$ is a ring homomorphism and M is a finitely generated A-module, then $\mathrm{Supp}(B \otimes_A M) = f^{*-1}(\mathrm{Supp}(M))$.

Solution. (i) If $M \neq 0$, then $M_\mathfrak{m} \neq 0$ for some maximal $\mathfrak{m}$ (3.8), so $\mathfrak{m} \in \mathrm{Supp}\, M$. Conversely $M = 0 \Rightarrow M_\mathfrak{p} = 0$ for all $\mathfrak{p}$.

(ii) $(A/\mathfrak{a})_\mathfrak{p} \cong A_\mathfrak{p}/\mathfrak{a}^e$ (3.4), which is $\neq 0$ iff $\mathfrak{a}^e \neq (1)$ iff $\mathfrak{a} \cap (A - \mathfrak{p}) = \emptyset$ (3.11(ii)) iff $\mathfrak{a} \subseteq \mathfrak{p}$.

(iii) Localization is exact: $0 \to M'_\mathfrak{p} \to M_\mathfrak{p} \to M''_\mathfrak{p} \to 0$. If $M_\mathfrak{p} = 0$ then $M'_\mathfrak{p}$ and $M''_\mathfrak{p}$ vanish; conversely if both vanish then so does $M_\mathfrak{p}$.

(iv) $(\sum M_i)_\mathfrak{p} = \sum (M_i)_\mathfrak{p}$ (3.4(i), valid for arbitrary sums, which are unions of finite sums). It is 0 iff every $(M_i)_\mathfrak{p} = 0$.

(v) If $\mathfrak{p} \in \mathrm{Supp}\, M$ and there were $s \in \mathrm{Ann}(M) - \mathfrak{p}$ then $M_\mathfrak{p} = 0$ (as $m/t = sm/st = 0$); so $\mathrm{Ann}\, M \subseteq \mathfrak{p}$. Conversely if $\mathrm{Ann}(M) \subseteq \mathfrak{p}$, then $\mathrm{Ann}(M_\mathfrak{p}) = (\mathrm{Ann}\, M)_\mathfrak{p} \subseteq \mathfrak{p}A_\mathfrak{p} \neq A_\mathfrak{p}$ by 3.14 (M f.g.), so $M_\mathfrak{p} \neq 0$.

(vi) By 3.7, $(M \otimes N)_\mathfrak{p} \cong M_\mathfrak{p} \otimes_{A_\mathfrak{p}} N_\mathfrak{p}$, and $M_\mathfrak{p}, N_\mathfrak{p}$ are finitely generated over the local ring $A_\mathfrak{p}$. By Ch. 2, Ex. 3 this is $\neq 0$ iff $M_\mathfrak{p} \neq 0$ and $N_\mathfrak{p} \neq 0$.

(vii) $M/\mathfrak{a}M \cong (A/\mathfrak{a})\otimes M$, with both factors finitely generated, so by (vi), (ii), (v): $\operatorname{Supp}(M/\mathfrak{a}M) = V(\mathfrak{a}) \cap V(\operatorname{Ann} M) = V(\mathfrak{a} + \operatorname{Ann} M)$.

(viii) Let $\mathfrak{q} \in \operatorname{Spec} B$, $\mathfrak{p} = f^{-1}(\mathfrak{q})$. $(B\otimes_A M)_\mathfrak{q} \cong B_\mathfrak{q}\otimes_A M \cong B_\mathfrak{q}\otimes_{A_\mathfrak{p}} M_\mathfrak{p}$ (note $B_\mathfrak{q}$ is an $A_\mathfrak{p}$-algebra, since $f(A-\mathfrak{p}) \subseteq B-\mathfrak{q}$, and $f(\mathfrak{p}) \subseteq \mathfrak{q}$). If $M_\mathfrak{p} = 0$, this is 0. If $M_\mathfrak{p} \neq 0$: $M_\mathfrak{p}$ is finitely generated over the local ring $A_\mathfrak{p}$, so $\bar{M} = M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p} = \kappa \otimes M_\mathfrak{p} \neq 0$ by Nakayama, where $\kappa = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$. Let $k(\mathfrak{q}) = B_\mathfrak{q}/\mathfrak{q}B_\mathfrak{q}$, a κ-algebra (because $f(\mathfrak{p}) \subseteq \mathfrak{q}$). The module $B_\mathfrak{q} \otimes_{A_\mathfrak{p}} M_\mathfrak{p}$ maps onto $k(\mathfrak{q})\otimes_{A_\mathfrak{p}} M_\mathfrak{p} \cong k(\mathfrak{q})\otimes_\kappa(\kappa\otimes_{A_\mathfrak{p}} M_\mathfrak{p}) = k(\mathfrak{q})\otimes_\kappa \bar{M}$, which is non-zero (tensor of two non-zero vector spaces over the field κ). So $(B\otimes M)_\mathfrak{q} \neq 0$. Hence $\mathfrak{q} \in \operatorname{Supp}(B\otimes M) \iff f^*(\mathfrak{q}) \in \operatorname{Supp}(M)$. □

Exercise 20

Problem. Let $f : A \to B$ be a ring homomorphism, $f^* : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ the associated mapping. Show that

(i) Every prime ideal of A is a contracted ideal $\iff$ f^* is surjective.

(ii) Every prime ideal of B is an extended ideal $\Rightarrow$ f^* is injective.

Is the converse of (ii) true?

Solution. (i) By 1.17(iii) a prime $\mathfrak{p}$ of A is a contracted ideal iff $\mathfrak{p}^{ec} = \mathfrak{p}$, which by 3.16 holds iff $\mathfrak{p}$ is the contraction of a prime of B, i.e. iff $\mathfrak{p} \in f^*(\operatorname{Spec} B)$.

(ii) Suppose all primes of B are extended. If $\mathfrak{q}$ is extended, $\mathfrak{q}^{ce} = \mathfrak{q}$ (1.17). Let $\mathfrak{q}_1, \mathfrak{q}_2$ be primes with $\mathfrak{q}_1^c = \mathfrak{q}_2^c$. Then $\mathfrak{q}_1 = (\mathfrak{q}_1^c)^e = (\mathfrak{q}_2^c)^e = \mathfrak{q}_2$.

The converse of (ii) is false. Let k be a field, $A = k$ and $B = k[t]/(t^2)$, f the inclusion. $\operatorname{Spec} B$ is a single point (its only prime is (t)), so f^* is injective. But the only extended ideals of B are $0^e = 0$ and $k^e = B$, so the prime (t) is not an extended ideal. □

Exercise 21

Problem.

(i) Let A be a ring, S a multiplicatively closed subset of A, and $\phi : A \to S^{-1}A$ the canonical homomorphism. Show that $\phi^* : \operatorname{Spec}(S^{-1}A) \to \operatorname{Spec}(A)$ is a homeomorphism of $\operatorname{Spec}(S^{-1}A)$ onto its image in $X = \operatorname{Spec}(A)$. Let this image be denoted by $S^{-1}X$. In particular, if $f \in A$, the image of $\operatorname{Spec}(A_f)$ in X is the basic open set X_f (Chapter 1, Exercise 17).

(ii) Let $f : A \to B$ be a ring homomorphism. Let $X = \operatorname{Spec}(A)$ and $Y = \operatorname{Spec}(B)$, and let $f^* : Y \to X$ be the mapping associated with f. Identifying $\operatorname{Spec}(S^{-1}B)$ (= $\operatorname{Spec}(f(S)^{-1}B)$) with its canonical image $S^{-1}Y$ in Y, and $\operatorname{Spec}(S^{-1}A)$ with its image $S^{-1}X$ in X, show that $\operatorname{Spec}(S^{-1}f) : \operatorname{Spec}(S^{-1}B) \to \operatorname{Spec}(S^{-1}A)$ is the restriction of f^* to $S^{-1}Y$, and that $S^{-1}Y = f^{*-1}(S^{-1}X)$.

(iii) Let $\mathfrak{a}$ be an ideal of A and let $\mathfrak{b} = \mathfrak{a}^e$ be its extension in B. Let $\bar{f} : A/\mathfrak{a} \to B/\mathfrak{b}$ be the homomorphism induced by f. If $\operatorname{Spec}(A/\mathfrak{a})$ is identified with its canonical image $V(\mathfrak{a})$ in X, and $\operatorname{Spec}(B/\mathfrak{b})$ with its image $V(\mathfrak{b})$ in Y, show that $\bar{f}^*$ is the restriction of f^* to $V(\mathfrak{b})$.

(iv) Let $\mathfrak{p}$ be a prime ideal of A. Take $S = A - \mathfrak{p}$ in (ii) and then reduce mod $S^{-1}\mathfrak{p}$ as in (iii). Deduce that the subspace $f^{*-1}(\mathfrak{p})$ of Y is naturally homeomorphic to $\mathrm{Spec}(B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}) = \mathrm{Spec}(k(\mathfrak{p}) \otimes_A B)$, where $k(\mathfrak{p})$ is the residue field of the local ring $A_\mathfrak{p}$. $\mathrm{Spec}(k(\mathfrak{p}) \otimes_A B)$ is called the *fiber* of f^* over $\mathfrak{p}$.

Solution. (i) By 3.11(iv), ϕ^* is injective with image $S^{-1}X = \{\mathfrak{p} : \mathfrak{p} \cap S = \emptyset\}$ ($\mathfrak{q} \mapsto \mathfrak{q}^c$, inverse $\mathfrak{p} \mapsto \mathfrak{p}^e = S^{-1}\mathfrak{p}$). It is continuous (Ch. 1, Ex. 21). Closed sets of $\mathrm{Spec}(S^{-1}A)$ are $V(\mathfrak{b})$ with $\mathfrak{b} = \mathfrak{a}^e$ for some ideal $\mathfrak{a}$ of A (3.11(i)). Now $\phi^*(V(\mathfrak{a}^e)) = V(\mathfrak{a}) \cap S^{-1}X$: if $\mathfrak{q} \supseteq \mathfrak{a}^e$ then $\mathfrak{q}^c \supseteq \mathfrak{a}$; conversely if $\mathfrak{p} \in S^{-1}X$, $\mathfrak{p} \supseteq \mathfrak{a}$, then $\mathfrak{p}^e \supseteq \mathfrak{a}^e$ and $\mathfrak{p} = (\mathfrak{p}^e)^c$. So ϕ^* maps closed sets onto the closed sets of the subspace $S^{-1}X$; thus it is a homeomorphism onto its image. For $S = \{f^n\}$: $S^{-1}X = \{\mathfrak{p} : f \notin \mathfrak{p}\} = X_f$.

(ii) For $\mathfrak{q} \in Y$: $\mathfrak{q} \cap f(S) = \emptyset \iff f(s) \notin \mathfrak{q}\ \forall s \iff s \notin f^{-1}(\mathfrak{q})\ \forall s \iff f^*(\mathfrak{q}) \cap S = \emptyset$. So $S^{-1}Y = f^{*-1}(S^{-1}X)$. Let $\phi_A : A \to S^{-1}A$, $\phi_B : B \to S^{-1}B$ be canonical, so $S^{-1}f \circ \phi_A = \phi_B \circ f$ (with $S^{-1}B \cong f(S)^{-1}B$, Ex. 4). Then $\phi_A^* \circ (S^{-1}f)^* = f^* \circ \phi_B^*$. Under the identifications by ϕ_A^*, ϕ_B^* (homeomorphisms onto $S^{-1}X$, $S^{-1}Y$ by (i)) this says $(S^{-1}f)^*$ is the restriction of f^* to $S^{-1}Y$.

(iii) Let $\pi_A : A \to A/\mathfrak{a}$, $\pi_B : B \to B/\mathfrak{b}$; then $\bar{f}\pi_A = \pi_B f$ (well defined because $f(\mathfrak{a}) \subseteq \mathfrak{b}$). Dually $\pi_A^*\bar{f}^* = f^*\pi_B^*$. By Ch. 1, Ex. 21(iv), π_A^*, π_B^* are homeomorphisms onto $V(\mathfrak{a})$, $V(\mathfrak{b})$; so $\bar{f}^*$ is the restriction of f^* to $V(\mathfrak{b})$ (and $f^*(V(\mathfrak{b})) \subseteq V(\mathfrak{a})$).

(iv) Take $S = A - \mathfrak{p}$ and $f_\mathfrak{p} = S^{-1}f : A_\mathfrak{p} \to B_\mathfrak{p}$. By (ii) the fibre of $f_\mathfrak{p}^*$ over the closed point $\mathfrak{p}A_\mathfrak{p}$ of $\mathrm{Spec}\, A_\mathfrak{p}$ is identified with $f^{*-1}(\mathfrak{p}) \subseteq S^{-1}Y$ (note $f^{*-1}(\mathfrak{p}) \subseteq S^{-1}Y$ since $\mathfrak{p} \cap S = \emptyset$). A prime Q of $B_\mathfrak{p}$ lies over $\mathfrak{p}A_\mathfrak{p}$ iff $Q^c = \mathfrak{p}A_\mathfrak{p}$ iff $Q \supseteq (\mathfrak{p}A_\mathfrak{p})^e = \mathfrak{p}B_\mathfrak{p}$ (since $\mathfrak{p}A_\mathfrak{p}$ is maximal). So the fibre is $V(\mathfrak{p}B_\mathfrak{p}) \cong \mathrm{Spec}(B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p})$ by (iii) (Ch. 1, Ex. 21(iv)); and the topology is the subspace topology throughout. Finally $B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p} \cong (A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}) \otimes_{A_\mathfrak{p}} B_\mathfrak{p} = k(\mathfrak{p}) \otimes_{A_\mathfrak{p}} (A_\mathfrak{p} \otimes_A B) \cong k(\mathfrak{p}) \otimes_A B$. □

Exercise 22

Problem. Let A be a ring and $\mathfrak{p}$ a prime ideal of A. Then the canonical image of $\mathrm{Spec}(A_\mathfrak{p})$ in $\mathrm{Spec}(A)$ is equal to the intersection of all the open neighborhoods of $\mathfrak{p}$ in $\mathrm{Spec}(A)$.

Solution. By 3.13 (or Ex. 21(i)) the image is $\{\mathfrak{q} : \mathfrak{q} \subseteq \mathfrak{p}\}$. Every open neighbourhood of $\mathfrak{p}$ contains a basic open set $X_f \ni \mathfrak{p}$, i.e. with $f \notin \mathfrak{p}$ (Ch. 1, Ex. 17); so the intersection of all neighbourhoods is $\bigcap_{f \notin \mathfrak{p}} X_f = \{\mathfrak{q} :\ f \notin \mathfrak{q} \text{ for all } f \notin \mathfrak{p}\}$. And "$f \notin \mathfrak{p} \Rightarrow f \notin \mathfrak{q}$" for all f is equivalent to $\mathfrak{q} \subseteq \mathfrak{p}$ (contrapositive: $f \in \mathfrak{q} \Rightarrow f \in \mathfrak{p}$). □

Exercise 23

Problem. Let A be a ring, let $X = \mathrm{Spec}(A)$ and let U be a basic open set in X (i.e., $U = X_f$ for some $f \in A$: Chapter 1, Exercise 17).

(i) If $U = X_f$, show that the ring $A(U) = A_f$ depends only on U and not on f.

(ii) Let $U' = X_g$ be another basic open set such that $U' \subseteq U$. Show that there is an equation of the form $g^n = uf$ for some integer $n > 0$ and some $u \in A$, and use this to define a homomorphism $\rho : A(U) \to A(U')$ (i.e., $A_f \to A_g$) by mapping a/f^m to au^m/g^{mn}. Show that ρ depends only on U and U'. This homomorphism is called the *restriction* homomorphism.

(iii) If $U = U'$, then ρ is the identity map.

(iv) If $U \supseteq U' \supseteq U''$ are basic open sets in X, show that the triangle of restriction homomorphisms commutes: $\rho_{UU''} = \rho_{U'U''} \circ \rho_{UU'}$, where $\rho_{UU''} : A(U) \to A(U'')$ is the direct arrow and the other two compose through $A(U')$.

(v) Let x $(= \mathfrak{p})$ be a point of X. Show that $\varinjlim_{U \ni x} A(U) \cong A_{\mathfrak{p}}$.

The assignment of the ring $A(U)$ to each basic open set U of X, and the restriction homomorphisms ρ, satisfying the conditions (iii) and (iv) above, constitutes a *presheaf of rings* on the basis of open sets $(X_f)_{f \in A}$. (v) says that the stalk of this presheaf at $x \in X$ is the corresponding local ring $A_{\mathfrak{p}}$.

Solution. *Key fact.* For $f, g \in A$: $X_g \subseteq X_f \iff V(f) \subseteq V(g) \iff g \in \bigcap_{\mathfrak{p} \ni f} \mathfrak{p} = r((f)) \iff g^n \in (f)$ for some $n > 0$ (1.14).

(i) Let $X_f = X_g$. Then $g^n = uf$ and $f^m = vg$ for some n, m, u, v. In A_f the element f is a unit, and $f^m = vg$ gives $g \cdot (vf^{-m}) = 1$, so g is a unit of A_f; symmetrically f is a unit of A_g. By 3.1 the canonical maps $A \to A_f, A_g$ induce A-algebra homomorphisms $A_f \to A_g \to A_f$ whose composite is the identity (uniqueness in 3.1). So $A_f \cong A_g$ by a unique A-algebra isomorphism, i.e. $A(U)$ is determined by U up to unique canonical isomorphism.

(ii) $U' \subseteq U$ gives $g^n = uf$ by the key fact. Then $f \cdot u = g^n$ is a unit in A_g, so f is a unit of A_g and by 3.1 there is a unique A-algebra homomorphism $\rho : A_f \to A_g$, with $\rho(a/f^m) = (a/1)(f/1)^{-m}$. In A_g, $f^{-1} = u\, g^{-n}$, so $\rho(a/f^m) = au^m/g^{mn}$, as stated. Because ρ is characterized as the unique A-algebra homomorphism $A_f \to A_g$, it does not depend on the choices of n, u (nor, by (i), on f, g beyond U, U').

(iii) If $U = U'$ the identity is an A-algebra homomorphism $A(U) \to A(U)$, so $\rho = \mathrm{id}$ by uniqueness.

(iv) $\rho_{U'U''} \circ \rho_{UU'}$ and $\rho_{UU''}$ are both A-algebra homomorphisms $A(U) \to A(U'')$; by uniqueness (3.1) they are equal.

(v) The neighbourhoods $U = X_f \ni \mathfrak{p}$ (i.e. $f \notin \mathfrak{p}$) form a directed system $(X_f \cap X_{f'} = X_{ff'})$, with the maps ρ. For $f \notin \mathfrak{p}$ the canonical $A_f \to A_{\mathfrak{p}}$ exists by 3.1 and is compatible with the ρ's (all are A-algebra homomorphisms), so by Ch. 2, Ex. 16 there is $\lambda : \varinjlim A(U) \to A_{\mathfrak{p}}$. It is surjective: $a/s \in A_{\mathfrak{p}}$ $(s \notin \mathfrak{p})$ is the image of $a/s \in A_s$. It is injective: an element of the limit is the class of some $\xi = a/f^m \in A_f$ (Ch. 2, Ex. 15), and $\lambda(\xi) = 0$ means $a/f^m = 0$ in $A_{\mathfrak{p}}$, so $sa = 0$ for some $s \notin \mathfrak{p}$. Then in $A_{fs} = A(X_f \cap X_s)$, with $X_{fs} \ni \mathfrak{p}$, the image of ξ is $a/f^m = 0$ (s is a unit there and $sa = 0$). So ξ is 0 in the limit. Hence $\varinjlim_{U \ni x} A(U) \cong A_{\mathfrak{p}}$. □

Exercise 24

Problem. Show that the presheaf of Exercise 23 has the following property. Let $(U_i)_{i \in I}$ be a covering of X by basic open sets. For each $i \in I$ let $s_i \in A(U_i)$ be such that, for each pair of indices i, j, the images of s_i and s_j in $A(U_i \cap U_j)$ are equal. Then there exists a unique $s \in A$ $(= A(X))$ whose image in $A(U_i)$ is s_i, for all $i \in I$. (This essentially implies that the presheaf is a *sheaf*.)

Solution. Write $U_i = X_{f_i}$, so $s_i \in A_{f_i}$.

Finiteness. $\bigcup X_{f_i} = X$ means no prime contains all f_i, so the f_i generate (1): $1 = \sum_{i\in J} g_i f_i$ with $J \subseteq I$ finite (Ch. 1, Ex. 17(v)). Raising to powers, for every N the elements f_i^N $(i \in J)$ also generate (1) (otherwise a maximal ideal would contain all f_i).

Normalizing. For $i \in J$ write $s_i = b_i/f_i^n$ with a common n. The hypothesis says $b_i/f_i^n = b_j/f_j^n$ in $A_{f_if_j}$, so $(f_if_j)^m(b_if_j^n - b_jf_i^n) = 0$ in A for some m (taking the maximum over the finitely many pairs). Put $h_i = f_i^{n+m}$ and $b_i' = b_if_i^m$; then $s_i = b_i'/h_i$ in A_{f_i} and $b_i'h_j - b_j'h_i = (f_if_j)^m(b_if_j^n - b_jf_i^n) = 0$ for $i, j \in J$. Choose g_i' with $\sum_{i\in J} g_i'h_i = 1$.

Existence. Let $s = \sum_{j\in J} g_j'b_j' \in A$. For $i \in J$:

$$s\,h_i = \sum_j g_j'b_j'h_i = \sum_j g_j'b_i'h_j = b_i' \sum_j g_j'h_j = b_i',$$

so the image of s in A_{f_i} is $b_i'/h_i = s_i$. Now let $i \in I$ be arbitrary and put $t = s_i - s \in A_{f_i}$. For $j \in J$ the image of t in $A_{f_if_j}$ is $(s_i)| - (s)| = (s_j)| - (s)| = 0$ by the hypothesis and the previous step. Since $A_{f_if_j} = (A_{f_i})_{f_j}$, there is k with $f_j^k t = 0$ in A_{f_i} (for all $j \in J$, with a common k). As the f_j^k $(j \in J)$ generate (1), $t = 0$. So $s \mapsto s_i$ for every $i \in I$.

Uniqueness. If $s, s' \in A$ both map to s_i for all i, then $t = s - s'$ maps to 0 in each A_{f_i}, so for $i \in J$ there is k with $f_i^k t = 0$ in A; the f_i^k generate (1), so $t = 0$. □

Exercise 25

Problem. Let $f : A \to B$, $g : A \to C$ be ring homomorphisms and let $h : A \to B \otimes_A C$ be defined by $h(x) = f(x) \otimes g(x)$. Let X, Y, Z, T be the prime spectra of $A, B, C, B \otimes_A C$ respectively. Then $h^*(T) = f^*(Y) \cap g^*(Z)$. [Let $\mathfrak{p} \in X$, and let $k = k(\mathfrak{p})$ be the residue field at $\mathfrak{p}$. By Exercise 21, the fiber $h^{*-1}(\mathfrak{p})$ is the spectrum of $(B \otimes_A C) \otimes_A k \cong (B \otimes_A k) \otimes_k (C \otimes_A k)$. Hence $\mathfrak{p} \in h^*(T) \iff (B \otimes_A k) \otimes_k (C \otimes_A k) \neq 0 \iff B \otimes_A k \neq 0$ and $C \otimes_A k \neq 0 \iff \mathfrak{p} \in f^*(Y) \cap g^*(Z)$.]

Solution. *Remark on the formula for h.* In $B \otimes_A C$ one has $f(x) \otimes 1 = 1 \otimes g(x) = x \cdot (1 \otimes 1)$, so $f(x) \otimes g(x) = x^2(1 \otimes 1)$, which is not additive in general (e.g. $(x+y)^2 \neq x^2 + y^2$ in $A = \mathbf{Z}$ with $B = C = \mathbf{Z}$). The same formula is printed in the text on p. 31. The intended homomorphism is $h(x) = f(x) \otimes 1 = 1 \otimes g(x)$ (this is the structure map of the A-algebra $B \otimes_A C$), and that is what I use.

Let $\mathfrak{p} \in X$, $k = k(\mathfrak{p})$. By Ex. 21(iv) the fibres are

$$h^{*-1}(\mathfrak{p}) = \operatorname{Spec}\big((B \otimes_A C) \otimes_A k\big), \quad f^{*-1}(\mathfrak{p}) = \operatorname{Spec}(B \otimes_A k), \quad g^{*-1}(\mathfrak{p}) = \operatorname{Spec}(C \otimes_A k).$$

A spectrum is non-empty iff the ring is non-zero. By the canonical isomorphisms of Ch. 2,

$$(B \otimes_A C) \otimes_A k \cong B \otimes_A k \otimes_A C \cong (B \otimes_A k) \otimes_k (C \otimes_A k),$$

a tensor product of two k-algebras, i.e. of two k-vector spaces, over the field k; it is non-zero iff both factors are non-zero ($\dim(V \otimes_k W) = \dim V \dim W$). Hence $\mathfrak{p} \in h^*(T) \iff B \otimes_A k \neq 0$ and $C \otimes_A k \neq 0 \iff \mathfrak{p} \in f^*(Y) \cap g^*(Z)$. □

Exercise 26

Problem. Let $(B_\alpha, g_{\alpha\beta})$ be a direct system of rings and B the direct limit. For each α, let $f_\alpha : A \to B_\alpha$ be a ring homomorphism such that $g_{\alpha\beta} \circ f_\alpha = f_\beta$ whenever $\alpha \leq \beta$ (i.e. the B_α form a direct system of A-algebras). The f_α induce $f : A \to B$. Show that

$$f^*(\mathrm{Spec}(B)) = \bigcap_\alpha f_\alpha^*(\mathrm{Spec}(B_\alpha)).$$

[Let $\mathfrak{p} \in \mathrm{Spec}(A)$. Then $f^{*-1}(\mathfrak{p})$ is the spectrum of $B \otimes_A k(\mathfrak{p}) \cong \varinjlim(B_\alpha \otimes_A k(\mathfrak{p}))$ (since tensor products commute with direct limits: Chapter 2, Exercise 20). By Exercise 21 of Chapter 2 it follows that $f^{*-1}(\mathfrak{p}) = \emptyset$ if and only if $B_\alpha \otimes_A k(\mathfrak{p}) = 0$ for some α, i.e. if and only if $f_\alpha^{*-1}(\mathfrak{p}) = \emptyset$.]

Solution. Let $g_\alpha : B_\alpha \to B$ be the canonical maps, so $f = g_\alpha f_\alpha$ and $f^* = f_\alpha^* g_\alpha^*$; hence $f^*(\mathrm{Spec}\, B) \subseteq f_\alpha^*(\mathrm{Spec}\, B_\alpha)$ for all α, giving $\subseteq$.

For $\supseteq$ let $\mathfrak{p} \in \bigcap_\alpha f_\alpha^*(\mathrm{Spec}\, B_\alpha)$, $k = k(\mathfrak{p})$. By Ex. 21(iv), $f_\alpha^{*-1}(\mathfrak{p}) = \mathrm{Spec}(B_\alpha \otimes_A k) \neq \emptyset$, i.e. $B_\alpha \otimes_A k \neq 0$ for every α. The ring homomorphisms $B_\alpha \otimes_A k \to B \otimes_A k$ induce a ring homomorphism $\varinjlim(B_\alpha \otimes_A k) \to B \otimes_A k$, bijective by Ch. 2, Ex. 20; so it is a ring isomorphism. If $B \otimes_A k$ were 0, then $\varinjlim(B_\alpha \otimes_A k) = 0$ and by Ch. 2, Ex. 21 some $B_\alpha \otimes_A k = 0$, a contradiction. So $B \otimes_A k \neq 0$, i.e. $f^{*-1}(\mathfrak{p}) \neq \emptyset$ and $\mathfrak{p} \in f^*(\mathrm{Spec}\, B)$. □

Exercise 27

Problem.

(i) Let $f_\alpha : A \to B_\alpha$ be any family of A-algebras and let $f : A \to B$ be their tensor product over A (Chapter 2, Exercise 23). Then $f^*(\mathrm{Spec}(B)) = \bigcap_\alpha f_\alpha^*(\mathrm{Spec}(B_\alpha))$. [Use Examples 25 and 26.]

(ii) Let $f_\alpha : A \to B_\alpha$ be any finite family of A-algebras and let $B = \prod_\alpha B_\alpha$. Define $f : A \to B$ by $f(x) = (f_\alpha(x))$. Then $f^*(\mathrm{Spec}(B)) = \bigcup_\alpha f_\alpha^*(\mathrm{Spec}(B_\alpha))$.

(iii) Hence the subsets of $X = \mathrm{Spec}(A)$ of the form $f^*(\mathrm{Spec}(B))$, where $f : A \to B$ is a ring homomorphism, satisfy the axioms for closed sets in a topological space. The associated topology is the *constructible topology* on X. It is finer than the Zariski topology (i.e., there are more open sets, or equivalently more closed sets).

(iv) Let X_C denote the set X endowed with the constructible topology. Show that X_C is quasi-compact.

Solution. (i) $B = \varinjlim_J B_J$, where J runs over the finite subsets of the index set and $B_J = \bigotimes_{\alpha \in J} B_\alpha$ (Ch. 2, Ex. 23). Let $f_J : A \to B_J$. By Ex. 26, $f^*(\mathrm{Spec}\, B) = \bigcap_J f_J^*(\mathrm{Spec}\, B_J)$. By Ex. 25 and induction on $|J|$ (with $B_{J\cup\{\alpha\}} = B_J \otimes_A B_\alpha$), $f_J^*(\mathrm{Spec}\, B_J) = \bigcap_{\alpha\in J} f_\alpha^*(\mathrm{Spec}\, B_\alpha)$ (for $J = \emptyset$, $B_\emptyset = A$ and the image is X). Since every α lies in some J, $\bigcap_J \bigcap_{\alpha\in J} = \bigcap_\alpha$.

(ii) $\mathrm{Spec}(\prod B_\alpha) = \bigsqcup \mathrm{Spec}\, B_\alpha$ (Ch. 1, Ex. 22): the primes of B are $\pi_\alpha^{-1}(\mathfrak{q})$ for $\mathfrak{q} \in \mathrm{Spec}\, B_\alpha$, $\pi_\alpha : B \to B_\alpha$ the projection. And $f^{-1}(\pi_\alpha^{-1}\mathfrak{q}) = f_\alpha^{-1}(\mathfrak{q})$ since $\pi_\alpha f = f_\alpha$. So $f^*(\mathrm{Spec}\, B) = \bigcup_\alpha f_\alpha^*(\mathrm{Spec}\, B_\alpha)$.

(iii) $\emptyset = f^*(\mathrm{Spec}\, 0)$ (for $A \to 0$) and $X = f^*(\mathrm{Spec}\, A)$ (identity). Arbitrary intersections are of the required form by (i), finite unions by (ii). So these sets are the closed sets of a topology. Every Zariski closed set $V(\mathfrak{a}) = f^*(\mathrm{Spec}(A/\mathfrak{a}))$ (Ch. 1, Ex. 21(iv)) is constructible-closed; so the constructible topology is finer.

(iv) The closed sets of X_C are exactly the sets $f^*(\operatorname{Spec} B)$, so quasi-compactness means: any family $E_\alpha = f_\alpha^*(\operatorname{Spec} B_\alpha)$ of closed sets with the finite intersection property has non-empty intersection. By (i), $\bigcap E_\alpha = f^*(\operatorname{Spec} B)$ with $B = \varinjlim B_J$. If $\bigcap E_\alpha = \emptyset$, then $B = 0$, so by Ch. 2, Ex. 21 some $B_J = 0$ and then $\bigcap_{\alpha \in J} E_\alpha = f_J^*(\operatorname{Spec} B_J) = \emptyset$, contradicting the finite intersection property. □

Exercise 28

Problem. (Continuation of Exercise 27.)

(i) For each $g \in A$, the set X_g (Chapter 1, Exercise 17) is both open and closed in the constructible topology.

(ii) Let C' denote the smallest topology on X for which the sets X_g are both open and closed, and let $X_{C'}$ denote the set X endowed with this topology. Show that $X_{C'}$ is Hausdorff.

(iii) Deduce that the identity mapping $X_C \to X_{C'}$ is a homeomorphism. Hence a subset E of X is of the form $f^*(\operatorname{Spec}(B))$ for some $f : A \to B$ if and only if it is closed in the topology C'.

(iv) The topological space X_C is compact, Hausdorff and totally disconnected.

Solution. (i) X_g is Zariski open, hence open in the finer constructible topology. It is also closed there: $X_g = f^*(\operatorname{Spec} A_g)$ for $f : A \to A_g$ (Ex. 21(i)).

(ii) Let $\mathfrak{p} \neq \mathfrak{q}$ be primes. Then there is g in one but not the other, say $g \in \mathfrak{p} - \mathfrak{q}$ (or $\mathfrak{q} - \mathfrak{p}$). Then $\mathfrak{q} \in X_g$ and $\mathfrak{p} \in X \setminus X_g$; both X_g and its complement are open in C' (as X_g is clopen in C') and they are disjoint. So $X_{C'}$ is Hausdorff.

(iii) By (i), $C' \subseteq C$ (every C'-open set is C-open), so the identity $X_C \to X_{C'}$ is a continuous bijection. X_C is quasi-compact (Ex. 27(iv)) and $X_{C'}$ is Hausdorff by (ii); a continuous bijection from a compact space to a Hausdorff space is closed, hence a homeomorphism. So $C = C'$, and the closed sets of C' are exactly the sets $f^*(\operatorname{Spec} B)$.

(iv) $X_C = X_{C'}$ is quasi-compact (Ex. 27(iv)) and Hausdorff (ii), i.e. compact Hausdorff. For $\mathfrak{p} \neq \mathfrak{q}$ choose g as in (ii): X_g is clopen in C and contains exactly one of the two points. So no connected subset contains two points: X_C is totally disconnected. □

Exercise 29

Problem. Let $f : A \to B$ be a ring homomorphism. Show that $f^* : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is a continuous *closed* mapping (i.e., maps closed sets to closed sets) for the constructible topology.

Solution. *Closed.* A closed set of $Y_C = \operatorname{Spec}(B)_C$ is $F = h^*(\operatorname{Spec} D')$ with $h : B \to D'$. Then $f^*(F) = f^*h^*(\operatorname{Spec} D') = (h \circ f)^*(\operatorname{Spec} D')$ is closed in X_C.

Continuous. Let $E = h^*(\operatorname{Spec} D)$ be closed in X_C, $h : A \to D$. Put $h' : B \to B \otimes_A D$, $y \mapsto y \otimes 1$. We claim $f^{*-1}(E) = h'^*(\operatorname{Spec}(B \otimes_A D))$, which is closed in Y_C. For $\mathfrak{q} \in \operatorname{Spec} B$, $\mathfrak{p} = f^*(\mathfrak{q})$:

$$\mathfrak{q} \in h'^*(\operatorname{Spec}(B \otimes_A D)) \iff k(\mathfrak{q}) \otimes_B (B \otimes_A D) \neq 0 \iff k(\mathfrak{q}) \otimes_A D \neq 0$$

(Ex. 21(iv) for h'), and $k(\mathfrak{q}) \otimes_A D \cong k(\mathfrak{q}) \otimes_{k(\mathfrak{p})} \big(k(\mathfrak{p}) \otimes_A D\big)$ because $k(\mathfrak{q})$ is a field extension of $k(\mathfrak{p})$; this is $\neq 0$ iff $k(\mathfrak{p}) \otimes_A D \neq 0$ (a field extension is faithfully flat: $V \neq 0 \Rightarrow k(\mathfrak{q}) \otimes_{k(\mathfrak{p})} V \neq 0$), i.e. iff $\mathfrak{p} \in h^*(\operatorname{Spec} D) = E$ (Ex. 21(iv) for h). □

Exercise 30

> **Problem.** Show that the Zariski topology and the constructible topology on $\operatorname{Spec}(A)$ are the same if and only if $A/\mathfrak{N}$ is absolutely flat (where $\mathfrak{N}$ is the nilradical of A). [Use Exercise 11.]

Solution. ($\Leftarrow$) If $A/\mathfrak{N}$ is absolutely flat, $\operatorname{Spec} A$ with the Zariski topology is Hausdorff (Ex. 11). The identity $X_C \to X_Z$ is a continuous bijection (the constructible topology is finer, Ex. 27(iii)) from a compact space X_C (Ex. 27(iv)) to a Hausdorff space, hence a homeomorphism. So the two topologies agree.

($\Rightarrow$) If they agree, the Zariski topology equals X_C, which is Hausdorff by Ex. 28. Then $\operatorname{Spec} A$ is Hausdorff, so $A/\mathfrak{N}$ is absolutely flat by Ex. 11. □

Notes

All page references are to the Addison–Wesley (1969) edition scanned in this repository. Exercise 25: the formula for h as printed in the scan is $h(x) = f(x) \otimes g(x)$; as explained in the solution, the intended (and only meaningful) homomorphism is $h(x) = f(x) \otimes 1$. The remaining exercises are solved as stated.